

Proposition 4. *Suppose $K \geq 2$. Then, for $R > 1$ sufficiently close to 1,*

$$d_{\min}^2 \geq 2E[1 - \eta(R^{K-1} + R^{-K+1})]. \quad (\star)$$

Before proceeding with the proof, we recall the inner product formula [1, Eq. (13)]

$$I_{\mathbf{s}}(R) = \frac{1}{K+1} \sum_{\ell=0}^K A_{\mathbf{H}}(\zeta^\ell) P_{\mathbf{s}}(\zeta^\ell), \quad \zeta = e^{j2\pi/(K+1)}, \quad (1)$$

where

$$A_{\mathbf{H}}(\zeta^\ell) = E \left[1 - 2\eta \cos \left(2\pi \frac{\ell}{K+1} \right) \right], \quad \eta = \frac{1}{R^K + R^{-K}}, \quad (2)$$

$$P_{\mathbf{s}}(\zeta^\ell) = \prod_{k=0}^{K-1} B_k(\zeta^\ell)^{s_k}, \quad \mathbf{s} \in \{-1, 0, 1\}^K, \quad (3)$$

$$B_k(\zeta^\ell) = R^{-1} \frac{\zeta^\ell - R\phi^k}{\zeta^\ell - R^{-1}\phi^k}, \quad \phi = e^{j2\pi/K}. \quad (4)$$

Now for the proof.

Proof. Our approach is as follows:

- 1) develop a tight approximation for $|I_{\mathbf{s}}(R)|$ when $R > 1$ is close to 1;
- 2) show that the single-flip signed patterns $\{\pm \mathbf{e}_k : k \in [K]\}$ maximize this approximation;
- 3) simplify $2E - 2|I_{\mathbf{e}_0}(R)|$ to obtain the lower bound in $(\star)$.

Step 1) Let us first consider the change of variables

$$r = \log(R), \quad R > 1.$$

Note that $r > 0$, and $r \rightarrow 0^+$ as $R \rightarrow 1^+$. Further, $R = e^r$ so that $B_k(\zeta^\ell)$ in (4) becomes

$$B_k(\zeta^\ell) = e^{-r} \frac{\zeta^\ell - e^r \phi^k}{\zeta^\ell - e^{-r} \phi^k}. \quad (5)$$

Taking the principal logarithm of both sides of (5), for small $r > 0$ we obtain¹

$$\text{Log} B_k(\zeta^\ell) = -r + \text{Log}(\zeta^\ell - e^r \phi^k) - \text{Log}(\zeta^\ell - e^{-r} \phi^k). \quad (6)$$

We now develop a Taylor approximation for $\text{Log} B_k(\zeta^\ell)$ about $r = 0$. Before that, however, it is important to note that when $r = 0$ and $\ell = k = 0$, $\text{Log} B_0(1)|_{r=0} = \text{Log}(0)$ is undefined. Hence, the following approximations are derived assuming $\ell \neq 0$.

¹See Appendix A for a justification of the identity $\text{Log}(z_1/z_2) = \text{Log}(z_1) - \text{Log}(z_2)$ applied here.

Computing the derivatives of $\text{Log}B_k(\zeta^\ell)$ at $r = 0$, we have

$$\begin{aligned}\text{Log}B_k(\zeta^\ell)\Big|_{r=0} &= 0, \\ \frac{d}{dr}\text{Log}B_k(\zeta^\ell)\Big|_{r=0} &= -\frac{\zeta^\ell + \phi^k}{\zeta^\ell - \phi^k} \stackrel{(a)}{=} j \cot\left(\pi\left(\frac{\ell}{K+1} - \frac{k}{K}\right)\right), \\ \frac{d^2}{dr^2}\text{Log}B_k(\zeta^\ell)\Big|_{r=0} &= 0,\end{aligned}$$

where (a) follows from Euler's cotangent identity (see [here](#)). Thus

$$\text{Log}B_k(\zeta^\ell) = jr \cot(\psi_{\ell,k}) + O(r^3), \quad \psi_{\ell,k} := \pi\left(\frac{\ell}{K+1} - \frac{k}{K}\right),$$

and the principal logarithm of $P_s(\zeta^\ell)$ in (3) simplifies to²

$$\text{Log}P_s(\zeta^\ell) = \sum_{k=0}^{K-1} s_k \text{Log}B_k(\zeta^\ell) = jr \sum_{k=0}^{K-1} s_k \cot(\psi_{\ell,k}) + O(r^3). \quad (7)$$

Defining $S_\ell(\mathbf{s}) := \sum_{k=0}^{K-1} s_k \cot(\psi_{\ell,k})$ and exponentiating both sides of (7), we obtain

$$P_s(\zeta^\ell) = \exp(jr S_\ell(\mathbf{s}) + O(r^3)) \stackrel{(b)}{=} 1 + jr S_\ell(\mathbf{s}) - \frac{r^2}{2} S_\ell(\mathbf{s})^2 + O(r^3),$$

where (b) follows from the Taylor expansion of $\exp(\cdot)$. This approximation is valid for $\ell \neq 0$. If $\ell = 0$, $B_0(1) = -1$ for all $r > 0$, and

$$P_s(1) = \prod_{k=0}^{K-1} B_k(1)^{s_k} = (-1)^{s_0} \prod_{k=1}^{K-1} B_k(1)^{s_k} = (1 - 2s_0^2)(1 + O(r)) = 1 - 2s_0^2 + O(r),$$

since $B_k(1) = 1 + O(r)$ for each $k \neq 0$.

We now approximate $A_H(\zeta^\ell)$ near $r = 0$ by noting that [see (2)]

$$2\eta = \frac{2}{e^{Kr} + e^{-Kr}} = \text{sech}(Kr) \stackrel{(c)}{=} 1 - \frac{K^2}{2} r^2 + O(r^4),$$

where (c) follows from the Taylor expansion of $\text{sech}(\cdot)$. Hence, for unit energy³ $E = 1$ and $w_\ell := 1 - \cos(2\theta_\ell)$, $\theta_\ell = \pi\ell/(K+1)$, we have

$$A_H(\zeta^\ell) = 1 - 2\eta \cos(2\theta_\ell) = w_\ell + \frac{K^2}{2} r^2 (1 - w_\ell) + O(r^4).$$

²See Appendix A for a justification of the identity $\text{Log}(\prod_k z_k^{s_k}) = \sum_k s_k \text{Log}(z_k)$ applied here.

³The energy $E > 0$ simply scales $|I_s(R)|$, so we set $E = 1$ (without loss of generality).

For $\ell \neq 0$, the product $A_{\text{H}}(\zeta^\ell)P_{\text{s}}(\zeta^\ell)$ in (1) can thus be written as

$$\begin{aligned} A_{\text{H}}(\zeta^\ell)P_{\text{s}}(\zeta^\ell) &= \left[w_\ell + \frac{K^2}{2}r^2(1 - w_\ell) + O(r^4) \right] \left[1 + \text{j}rS_\ell(\mathbf{s}) - \frac{r^2}{2}S_\ell(\mathbf{s})^2 + O(r^3) \right] \\ &= w_\ell + \text{j}rw_\ell S_\ell(\mathbf{s}) + \frac{K^2}{2}r^2(1 - w_\ell) - \frac{r^2}{2}w_\ell S_\ell(\mathbf{s})^2 + O(r^3). \end{aligned} \quad (8)$$

For $\ell = 0$, however, we get

$$A_{\text{H}}(1)P_{\text{s}}(1) = \left[\frac{K^2}{2}r^2 + O(r^4) \right] [1 - 2s_0^2 + O(r)] = (1 - 2s_0^2)\frac{K^2}{2}r^2 + O(r^3).$$

To obtain a small- r approximation of (1), we sum (8) over $\ell \in \{1, \dots, K\}$ and use

$$\begin{aligned} \sum_{\ell=1}^K w_\ell &= K + 1, \\ \sum_{\ell=1}^K (1 - w_\ell) &= -1, \\ \sum_{\ell=1}^K w_\ell S_\ell(\mathbf{s}) &\stackrel{\text{(d)}}{=} 0, \end{aligned}$$

yielding

$$\begin{aligned} |I_{\text{s}}(e^r)| &= \frac{1}{K+1} \left| \sum_{\ell=0}^K A_{\text{H}}(\zeta^\ell)P_{\text{s}}(\zeta^\ell) \right| \\ &= \frac{1}{K+1} \left| A_{\text{H}}(1)P_{\text{s}}(1) + \sum_{\ell=1}^K A_{\text{H}}(\zeta^\ell)P_{\text{s}}(\zeta^\ell) \right| \\ &= \frac{1}{K+1} \left| (1 - 2s_0^2)\frac{K^2}{2}r^2 + (K+1) - \frac{K^2}{2}r^2 - \frac{r^2}{2} \sum_{\ell=1}^K w_\ell S_\ell(\mathbf{s})^2 \right| + O(r^3) \\ &= \frac{1}{K+1} \left| K+1 - K^2 s_0^2 r^2 - \frac{r^2}{2} \sum_{\ell=1}^K w_\ell S_\ell(\mathbf{s})^2 \right| + O(r^3) \\ &= \frac{1}{K+1} \left| K+1 - \frac{r^2}{2} Q(\mathbf{s}) \right| + O(r^3), \quad Q(\mathbf{s}) := 2K^2 s_0^2 + \sum_{\ell=1}^K w_\ell S_\ell(\mathbf{s})^2. \end{aligned} \quad (9)$$

The identity in (d) is not obvious but we defer its proof to Appendix B.

Step 2) We now seek the signed patterns maximizing (9). To that end, we note that $Q(\mathbf{s})$ determines the deviation of $|I_{\text{s}}(e^r)|$ away from 1 [up to a bounded error $O(r^3)$]. Therefore, for sufficiently small $r > 0$, maximizing (9) reduces to minimizing $Q(\mathbf{s})$.

Expanding $Q(\mathbf{s})$, we have

$$\begin{aligned}
Q(\mathbf{s}) &= 2K^2 s_0^2 + \sum_{\ell=1}^K w_\ell \sum_{k,m=0}^{K-1} s_k s_m \cot(\psi_{\ell,k}) \cot(\psi_{\ell,m}) \\
&= 2K^2 s_0^2 + \sum_{k,m=0}^{K-1} s_k s_m \sum_{\ell=1}^K w_\ell \cot(\psi_{\ell,k}) \cot(\psi_{\ell,m}) \\
&= \sum_{k,m=0}^{K-1} s_k s_m M_{km}, \quad M_{km} := 2K^2 \delta_{k0} \delta_{m0} + \sum_{\ell=1}^K w_\ell \cot(\psi_{\ell,k}) \cot(\psi_{\ell,m}). \quad (10)
\end{aligned}$$

Hence $Q(\mathbf{s}) = \mathbf{s}^T \mathbf{M} \mathbf{s}$ is a quadratic form (see [here](#)) and $\mathbf{M} \in \mathbb{R}^{K \times K}$ is a symmetric matrix. Let us compute a general off-diagonal element M_{km} , $k \neq m$. To do this, we reexpress w_ℓ using Euler's cosine identity as

$$w_\ell = 1 - \cos(2\theta_\ell) = 1 - \frac{\zeta^\ell + \zeta^{-\ell}}{2} = -\frac{(\zeta^\ell - 1)^2}{2\zeta^\ell}.$$

Similarly, we reexpress $\cot(\psi_{\ell,k})$ and $\cot(\psi_{\ell,m})$ using Euler's cotangent identity as

$$\cot(\psi_{\ell,k}) = j \frac{\zeta^\ell + \phi^k}{\zeta^\ell - \phi^k} \quad \text{and} \quad \cot(\psi_{\ell,m}) = j \frac{\zeta^\ell + \phi^m}{\zeta^\ell - \phi^m}.$$

Then

$$\begin{aligned}
w_\ell \cot(\psi_{\ell,k}) \cot(\psi_{\ell,m}) &= \frac{(\zeta^\ell - 1)^2 (\zeta^\ell + \phi^k) (\zeta^\ell + \phi^m)}{2\zeta^\ell (\zeta^\ell - \phi^k) (\zeta^\ell - \phi^m)} \\
&\stackrel{(e)}{=} (\phi^k + \phi^m - 1) + \frac{\zeta^\ell}{2} + \frac{\zeta^{-\ell}}{2} \\
&\quad + \frac{(\phi^k + \phi^m)(\phi^k - 1)^2}{(\phi^k - \phi^m)(\zeta^\ell - \phi^k)} - \frac{(\phi^k + \phi^m)(\phi^m - 1)^2}{(\phi^k - \phi^m)(\zeta^\ell - \phi^m)}, \quad (11)
\end{aligned}$$

where (e) results from a partial fraction decomposition (which is unique).⁴ In light of

$$\sum_{\ell=1}^K \zeta^\ell = \sum_{\ell=1}^K \zeta^{-\ell} = -1 \quad \text{and} \quad \sum_{\ell=1}^K \frac{1}{\zeta^\ell - \phi^k} = -\frac{K}{\phi^k - 1}, \quad (12)$$

summing (11) over ℓ yields

$$\begin{aligned}
M_{km} &= \sum_{\ell=1}^K w_\ell \cot(\psi_{\ell,k}) \cot(\psi_{\ell,m}) \\
&= K(\phi^k + \phi^m - 1) - 1 + \left(\frac{\phi^k + \phi^m}{\phi^k - \phi^m} \right) \left[(\phi^k - 1)^2 \cdot -\frac{K}{\phi^k - 1} - (\phi^m - 1)^2 \cdot -\frac{K}{\phi^m - 1} \right] \\
&= K(\phi^k + \phi^m - 1) - 1 - K \left(\frac{\phi^k + \phi^m}{\phi^k - \phi^m} \right) (\phi^k - \phi^m) \\
&= K(\phi^k + \phi^m - 1) - 1 - K(\phi^k + \phi^m) \\
&= -(K + 1), \quad k \neq m. \quad (13)
\end{aligned}$$

⁴This decomposition can be computed in MATLAB using the `partfrac()` function (see Appendix C).

Now consider a diagonal element M_{kk} , $k \neq 0$. Using the previous Euler identities, we have

$$\begin{aligned} w_\ell \cot^2(\psi_{\ell,k}) &= \frac{(\zeta^\ell - 1)^2(\zeta^\ell + \phi^k)^2}{2\zeta^\ell(\zeta^\ell - \phi^k)^2} \\ &\stackrel{(f)}{=} (2\phi^k - 1) + \frac{\zeta^\ell}{2} + \frac{\zeta^{-\ell}}{2} + \frac{4\phi^k(\phi^k - 1)}{\zeta^\ell - \phi^k} + \frac{2\phi^k(\phi^k - 1)^2}{(\zeta^\ell - \phi^k)^2}, \end{aligned} \quad (14)$$

where (f) follows from a partial fraction decomposition. In light of (12) and

$$\sum_{\ell=1}^K \frac{1}{(\zeta^\ell - \phi^k)^2} = \frac{K(K+1+\phi^k)}{\phi^k(\phi^k - 1)^2},$$

summing (14) over ℓ yields

$$\begin{aligned} M_{kk} &= \sum_{\ell=1}^K w_\ell \cot^2(\psi_{\ell,k}) \\ &= K(2\phi^k - 1) - 1 + 4\phi^k(\phi^k - 1) \cdot -\frac{K}{\phi^k - 1} + 2\phi^k(\phi^k - 1)^2 \cdot \frac{K(K+1+\phi^k)}{\phi^k(\phi^k - 1)^2} \\ &= K(2\phi^k - 1) - 1 - 4K\phi^k + 2K(K+1+\phi^k) \\ &= -(K+1) + 2K(K+1) \\ &= (K+1)(2K-1), \quad k \neq 0. \end{aligned} \quad (15)$$

It only remains to compute M_{00} . To do this, we substitute $k = 0$ into (14) and sum over ℓ [while recalling (10) and the sum identities in (12)]:

$$M_{00} = 2K^2 + \sum_{\ell=1}^K w_\ell \cot^2(\psi_{\ell,0}) = 2K^2 + \frac{1}{2} \sum_{\ell=1}^K (2 + \zeta^\ell + \zeta^{-\ell}) = (K+1)(2K-1). \quad (16)$$

Combining (13), (15), and (16), we see that $\mathbf{M}$ has the simple form

$$M_{km} = \begin{cases} (K+1)(2K-1), & k = m \\ -(K+1), & k \neq m \end{cases}.$$

We can therefore simplify $Q(\mathbf{s})$ in (10) as

$$\begin{aligned} Q(\mathbf{s}) &= \sum_{k,m=0}^{K-1} s_k s_m M_{km} \\ &= (K+1)(2K-1) \sum_{k=m} s_k^2 - (K+1) \sum_{k \neq m} s_k s_m \\ &= (K+1) \left[2K \sum_{k=0}^{K-1} s_k^2 - \left(\sum_{k=0}^{K-1} s_k \right)^2 \right] \\ &= (K+1)[2Kv(\mathbf{s}) - u(\mathbf{s})^2], \end{aligned}$$

where $u(\mathbf{s}) := \sum_{k=0}^{K-1} s_k$ and $v(\mathbf{s}) := \sum_{k=0}^{K-1} s_k^2$. Because $|u(\mathbf{s})| \leq v(\mathbf{s})$ and $1 \leq v(\mathbf{s}) \leq K$,⁵ it holds that

$$Q(\mathbf{s}) \geq (K+1)[2Kv(\mathbf{s}) - v(\mathbf{s})^2] \geq (K+1)(2K-1).$$

We have equality if and only if $v(\mathbf{s}) = u(\mathbf{s})^2 = 1$, resulting when $\mathbf{s} = \pm \mathbf{e}_k$ for some $k \in [K]$. Thus, the single-flip patterns $\{\pm \mathbf{e}_k : k \in [K]\}$ maximize $|I_{\mathbf{s}}(e^r)|$ for sufficiently small $r > 0$.

Step 3) Plugging $\mathbf{s} = \mathbf{e}_0$ into (1) and simplifying (see Appendix D), we obtain

$$I_{\mathbf{e}_0}(R) = E\eta(R^{K-1} + R^{-K+1}). \quad (17)$$

The lower bound in $(\star)$ follows as

$$d_{\min}^2 \geq 2E - 2|I_{\mathbf{e}_0}(R)| = 2E[1 - \eta(R^{K-1} + R^{-K+1})],$$

which holds for $R > 1$ sufficiently close to 1. ■

Appendix A (Complex logarithm identities)

We first consider the identity in (6) for $\ell \neq 0$, which holds if and only if

$$\text{Arg}(\zeta^\ell - e^r \phi^k) - \text{Arg}(\zeta^\ell - e^{-r} \phi^k) \in (-\pi, \pi].$$

Towards verifying that the above condition holds, we make the following observation:

$$\lim_{r \rightarrow 0^+} (\zeta^\ell - e^r \phi^k) = \zeta^\ell - \phi^k = \lim_{r \rightarrow 0^+} (\zeta^\ell - e^{-r} \phi^k).$$

This observation is important, since if $\zeta^\ell - \phi^k \notin (-\infty, 0)$ (which implies $\text{Arg}(\zeta^\ell - \phi^k) \neq \pi$), it follows that $\text{Arg}(\zeta^\ell - e^r \phi^k) - \text{Arg}(\zeta^\ell - e^{-r} \phi^k) \approx 0 \in (-\pi, \pi]$ for sufficiently small $r > 0$. Hence, suppose $\zeta^\ell - \phi^k = x \in (-\infty, 0)$. Then

$$\begin{cases} \Re(\zeta^\ell - \phi^k) = \cos(2\pi\ell/(K+1)) - \cos(2\pi k/K) = x, \\ \Im(\zeta^\ell - \phi^k) = \sin(2\pi\ell/(K+1)) - \sin(2\pi k/K) = 0. \end{cases}$$

The latter equation requires

$$2\pi\ell/(K+1) \equiv 2\pi k/K \pmod{\pi} \iff \pi \mid 2\psi_{\ell,k} \iff 2\psi_{\ell,k} \in \{-\pi, 0, \pi\}.$$

Case $2\psi_{\ell,k} = 0$: This case results when $\ell = k = 0$. However, $\ell \neq 0$ by assumption, so we can ignore this case.

Case $2\psi_{\ell,k} = -\pi$: This case results for even K when $\ell = 0$ and $k = K/2$. Again, however, $\ell \neq 0$ by assumption, so we can ignore this case.

⁵The lower bound of one results since we exclude $\mathbf{s} = [0, \dots, 0]^T$ (which corresponds to a self distance).

Case $2\psi_{\ell,k} = \pi$: This case results for odd K when $\ell = (K+1)/2$ and $k = 0$. There

$$\cos(2\pi\ell/(K+1)) - \cos(2\pi k/K) = \cos(\pi) - 1 = -2 \in (-\infty, 0).$$

But for any $r > 0$, $\text{Arg}(-1 - e^r) - \text{Arg}(-1 - e^{-r}) = 0 \in (-\pi, \pi]$, so (6) holds in this case.

All other combinations of ℓ and k yield $\zeta^\ell - \phi^k \notin (-\infty, 0)$, in which case (6) holds.

We now consider the identity in (7) for $\ell \neq 0$, which holds if and only if

$$\sum_{k=0}^{K-1} s_k \text{Arg}(B_k(\zeta^\ell)) \in (-\pi, \pi].$$

In our case, for any $(\ell, k) \neq (0, 0)$, $B_k(\zeta^\ell) \rightarrow 1$ as $r \rightarrow 0^+$ so that $\text{Arg}(B_k(\zeta^\ell)) \rightarrow 0$. Hence $s_k \text{Arg}(B_k(\zeta^\ell)) \rightarrow 0$ and the above sum condition holds, at least for sufficiently small $r > 0$.

Appendix B (Cotangent sum identity)

We have

$$\sum_{\ell=1}^K w_\ell S_\ell(\mathbf{s}) = \sum_{\ell=1}^K w_\ell \sum_{k=0}^{K-1} s_k \cot(\psi_{\ell,k}) = \sum_{k=0}^{K-1} s_k \sum_{\ell=1}^K w_\ell \cot(\psi_{\ell,k}).$$

Now, as in the proof, we use Euler's cosine and cotangent identities to write, respectively,

$$w_\ell = -\frac{(\zeta^\ell - 1)^2}{2\zeta^\ell} \quad \text{and} \quad \cot(\psi_{\ell,k}) = j \frac{\zeta^\ell + \phi^k}{\zeta^\ell - \phi^k}.$$

Thus

$$w_\ell \cot(\psi_{\ell,k}) = -j \frac{(\zeta^\ell - 1)^2(\zeta^\ell + \phi^k)}{2\zeta^\ell(\zeta^\ell - \phi^k)} \stackrel{(h)}{=} -j \left[(\phi^k - 1) + \frac{\zeta^\ell}{2} - \frac{\zeta^\ell}{2} + \frac{(\phi^k - 1)^2}{\zeta^\ell - \phi^k} \right],$$

where (h) follows from a partial fraction decomposition. Summing over ℓ yields [see (12)]

$$\sum_{\ell=1}^K w_\ell \cot(\psi_{\ell,k}) = -j \left[K(\phi^k - 1) + (\phi^k - 1)^2 \cdot -\frac{K}{\phi^k - 1} \right] = 0,$$

and hence $\sum_{\ell=1}^K w_\ell S_\ell(\mathbf{s}) = 0$.

Appendix C (Partial fractions in MATLAB)

The MATLAB code below computes the partial fraction decomposition in (11) using the `partfrac()` function and the Symbolic Math Toolbox with $\mathbf{z} \equiv \zeta^\ell$, $\mathbf{a} \equiv \phi^k$, and $\mathbf{b} \equiv \phi^m$. Other decompositions can be computed similarly.

```
syms z a b
expression = (z-1)^2*(z+a)*(z+b) / (2*z*(z-a)*(z-b));
pretty(partfrac(expression, z))
```

Appendix D (Simplifying $I_{\mathbf{e}_0}(R)$)

Substituting $\mathbf{s} = \mathbf{e}_0$ into (1), only the $k = 0$ term $B_0(\zeta^\ell)$ in the product $P_{\mathbf{e}_0}(\zeta^\ell)$ contributes:

$$I_{\mathbf{e}_0}(R) = \frac{1}{K+1} \sum_{\ell=0}^K A_H(\zeta^\ell) R^{-1} \frac{\zeta^\ell - R}{\zeta^\ell - R^{-1}}.$$

Let us write

$$A_H(\zeta^\ell) = E[1 - \eta(\zeta^\ell + \zeta^{-\ell})] \quad \text{and} \quad \frac{\zeta^\ell - R}{\zeta^\ell - R^{-1}} = 1 + \frac{R^{-1} - R}{\zeta^\ell - R^{-1}}.$$

Then

$$\begin{aligned} I_{\mathbf{e}_0}(R) &= \frac{ER^{-1}}{K+1} \sum_{\ell=0}^K [1 - \eta(\zeta^\ell + \zeta^{-\ell})] \left(1 + \frac{R^{-1} - R}{\zeta^\ell - R^{-1}}\right) \\ &= \frac{ER^{-1}}{K+1} \left[\sum_{\ell=0}^K 1 + (R^{-1} - R) \sum_{\ell=0}^K \frac{1}{\zeta^\ell - R^{-1}} - \eta \sum_{\ell=0}^K (\zeta^\ell + \zeta^{-\ell}) \right. \\ &\quad \left. - \eta(R^{-1} - R) \sum_{\ell=0}^K \left(\frac{\zeta^\ell}{\zeta^\ell - R^{-1}} + \frac{\zeta^{-\ell}}{\zeta^\ell - R^{-1}} \right) \right]. \end{aligned}$$

Clearly $\sum_{\ell=0}^K 1 = K+1$ and $\sum_{\ell=0}^K (\zeta^\ell + \zeta^{-\ell}) = 0$ [see (12)]. Defining $q := R^{-1}$ and $N := K+1$, we also have the identities

$$\sum_{\ell=0}^{N-1} \frac{1}{\zeta^\ell - q} = \frac{Nq^{N-1}}{1 - q^N} \quad \text{and} \quad \sum_{\ell=0}^{N-1} \frac{\zeta^\ell}{\zeta^\ell - q} = \frac{N}{1 - q^N} \quad \text{and} \quad \sum_{\ell=0}^{N-1} \frac{\zeta^{-\ell}}{\zeta^\ell - q} = \frac{Nq^{N-2}}{1 - q^N}.$$

Thus

$$\begin{aligned} I_{\mathbf{e}_0}(q^{-1}) &= \frac{Eq}{N} \left[N + (q - q^{-1}) \left(\frac{Nq^{N-1}}{1 - q^N} - \eta \frac{N + Nq^{N-2}}{1 - q^N} \right) \right] \\ &\stackrel{(i)}{=} Eq \left[1 + (q - q^{-1}) \left(\frac{q^{N-1} - \frac{q^{N-1}}{1+q^{2N-2}}(1 + q^{N-2})}{1 - q^N} \right) \right], \end{aligned}$$

where in (i) we substituted $\eta = 1/(q^K + q^{-K}) = q^K/(1 + q^{2K}) = q^{N-1}/(1 + q^{2N-2})$ [see (2)]. Further algebra yields

$$\frac{q^{N-1} - \frac{q^{N-1}}{1+q^{2N-2}}(1 + q^{N-2})}{1 - q^N} = -\frac{q^{2N-3}}{1 + q^{2N-2}},$$

and hence

$$I_{\mathbf{e}_0}(q^{-1}) = Eq \left[1 - (q - q^{-1}) \frac{q^{2N-3}}{1 + q^{2N-2}} \right] = Eq \frac{1 + q^{2N-4}}{1 + q^{2N-2}} = E \frac{q^{-(N-2)} + q^{N-2}}{q^{-(N-1)} + q^{N-1}}.$$

Undoing our substitutions $q = R^{-1}$ and $N = K+1$, we obtain (17) as

$$I_{\mathbf{e}_0}(R) = E \frac{R^{K-1} + R^{-(K-1)}}{R^K + R^{-K}} = E\eta(R^{K-1} + R^{-K+1}).$$

References

- [1] P. Huggins, A. Şahin, and E. Erkip, “On the distance properties of Huffman sequences,” *IEEE Signal Process. Lett.*, pp. 1–5, under review.